

KALYAN BABU GUTTI

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PROFESSIONAL SUMMARY

Backend-focused Computer Science undergraduate specializing in Python and FastAPI, with hands-on experience building REST APIs, database-backed applications, and secure authentication systems using PostgreSQL, SQLAlchemy, and Redis. Experienced in integrating computer-vision and machine-learning models into backend services, with a strong foundation in data structures, algorithms, and software engineering fundamentals.

TECHNICAL SKILLS

Languages: Python, Java, SQL

Backend & APIs: FastAPI, RESTful API Design, SQLAlchemy, Asynchronous Programming, Request/Response Validation, Exception Handling

Databases & Caching: PostgreSQL, MySQL, Redis

Authentication & Security: JWT (Access/Refresh Tokens), OAuth2, Role-Based Access Control (RBAC), Password Hashing, Token Blacklisting

AI / Machine Learning: Machine Learning, Computer Vision, Scikit-learn, TensorFlow, NumPy, Pandas, OpenCV

Core CS Foundations: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, DBMS

Tools & Platforms: Git, GitHub, Postman

PROJECTS

Bike Taxi Backend API

Jul 2026 – Present

Python, FastAPI, SQLAlchemy, PostgreSQL, Redis

- Architected and built a modular, production-oriented backend using FastAPI, SQLAlchemy, and PostgreSQL with fully asynchronous database operations to support concurrent rider and driver workflows.
- Engineered secure RESTful APIs featuring JWT access/refresh token authentication, OAuth2 integration, bcrypt-based password hashing, role-based access control, and strict request/response validation.
- Implemented Redis-based token blacklisting to enable secure logout and session invalidation, reducing unauthorized token reuse risk across the platform.
- Structured the codebase into distinct service, database, security, and exception-handling layers for improved maintainability.
- GitHub:** github.com/KalyanGutti/Bike-Taxi

Retinal Disease Detection System

Jun 2025 – Dec 2025

Python, FastAPI, PostgreSQL, MobileNetV2

- Built and deployed a FastAPI backend integrated with PostgreSQL to serve real-time, image-based disease prediction requests and persist prediction history for downstream analysis.
- Designed RESTful endpoints for image upload, prediction inference, input validation, and structured error handling to ensure reliable API behavior under invalid or malformed inputs.
- Integrated a MobileNetV2 computer-vision model into the backend pipeline and implemented Grad-CAM visualizations to provide explainable, interpretable AI predictions for end users.
- GitHub:** github.com/kalyangutti/Retinal-Disease-Detection-System

EDUCATION

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

2023 – Present

B.Tech in Computer Science & Engineering | CGPA: 9.2/10

Narayana Junior College

2021 – 2023

Intermediate (Class XII) | Percentage: 96%

St. Ann's School

2021

10th ICSE | Percentage: 78%

CERTIFICATIONS

- Java Programming Fundamentals** | Infosys Springboard Jan 2025 – Mar 2025
- Data Structures & Algorithms in Python: Sorting Algorithms** | Infosys Springboard Nov 2024 – Jan 2025
- DBMS: Master the Fundamentals and Advanced Concepts** |Scaler Aug 2024 – Nov 2024

ACHIEVEMENTS & EXTRA-CURRICULAR ACTIVITIES

- Co-authored the research paper "Improving Nutritional Analysis with XGBoost," presented at ICICV 2025.
- Selected as a participant for the Internal Smart India Hackathon (SIH) 2025, gaining experience in rapid prototyping and structured problem-solving.